EXPERIMENT NO. 03– Simulation and Performance Analysis of a Poisson Arrival–Poisson Service Queue with Finite Buffer



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AIM: To simulate a communication system with Poisson packet arrivals and Poisson service times by implementing a finite buffer and evaluate the system performance by evaluating the packet drop probability.

INTRODUCTION:

In networks, queueing systems are used to analyse the network performance. In networks, packets arrive randomly, they are of random length. The main focus therefore shifts to delay related performance of the packets i.e- how long the packets would wait for and how much longer the buffer will be for the packets. This experiment tries to formulate a simulation so that the randomness can be captured for a network.

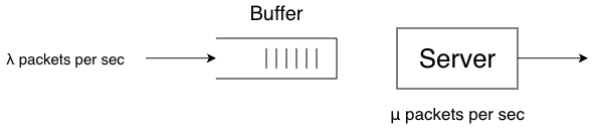


Figure 1: Buffer and Serving Process

THEORY:

The arrival process is modeled with Poisson process, which is a random process. In a Poisson process, the arrivals occur at an average rate of λ packets per second [1]. The probability of observing n arrivals in a time period T is given by

$$P(N(T) = n) = \frac{(\lambda T)^n e^{-\lambda T}}{n!}$$

The main idea in this experiment is to model the arrival process and service process based on Poisson process,. To do so, we use exponential random variable and the concept of inter arrival time. Inter arrival time denotes the time between two successive arrivals [1]. Following is the CDF of inter arrival time:

$$\begin{aligned} P(\text{IA} \leq t) &= 1 - P(\text{IA} > t) \\ &= 1 - P(0 \text{ arrivals in time } t) \\ &= 1 - e^{-\lambda t} \end{aligned}$$

The distribution is known as the **exponential distribution**.

So. inter-arrival CDF:

$$F_{\text{IA}}(t) = 1 - e^{-\lambda t}$$

And inter-arrival PDF:

$$f_{\text{IA}}(t) = \frac{d}{dt} F_{\text{IA}}(t) = \lambda e^{-\lambda t}$$

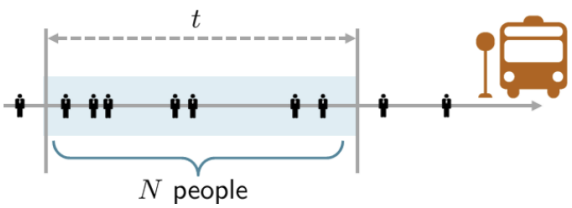


Figure 2: Inter arrival Process Example (Arrival of N people in between t time interval) [2]

Derivation of the Inter-arrival Time Distribution Let T denote the inter-arrival time between two consecutive events in a Poisson process with rate λ. Let t be the time period under observation [2].

$$\begin{aligned}
\mathbb{P}(T > t) &\overset{(a)}{=} \mathbb{P}(\text{inter-arrival time} > t) \\
&\overset{(b)}{=} \mathbb{P}(\text{no arrivals in time } t) \\
&\overset{(c)}{=} \mathbb{P}(N(t) = 0) \\
&= \frac{(\lambda t)^0}{0!} e^{-\lambda t} \\
&= e^{-\lambda t}.
\end{aligned}$$

Therefore, the cumulative distribution function (CDF) of the inter-arrival time is

$$F_{\text{IA}}(t) = \mathbb{P}(T \leq t) = 1 - e^{-\lambda t}, \quad t \geq 0.$$

Differentiating the CDF yields the probability density function (PDF):

$$f_{\text{IA}}(t) = \frac{d}{dt} F_{\text{IA}}(t) = \lambda e^{-\lambda t}, \quad t \geq 0.$$

Hence, the inter-arrival time $T = \text{IA}$ follows an exponential distribution with parameter λ.

RESULTS:

```
Simulation Time           : 10.00
Arrival rate (lambda)     : 10.00
Service rate (mu)         : 13.00
Buffer size (K)           : 5
---------------------------------
Total Arrivals            : 96
Total Serviced            : 83
Total Dropped             : 8
Packet Drop Probability   : 0.083333
---------------------------------
Storing Interaarival Times
Inter-arrival[1] = 0.143
Inter-arrival[2] = 0.143
Inter-arrival[3] = 0.119
Inter-arrival[4] = 0.046
```

```
Inter-arrival[5] = 0.123
Inter-arrival[6] = 0.020
Inter-arrival[7] = 0.142
Inter-arrival[8] = 0.172
Inter-arrival[9] = 0.214
Inter-arrival[10] = 0.161
Inter-arrival[11] = 0.011
Inter-arrival[12] = 0.038
Inter-arrival[13] = 0.312
Inter-arrival[14] = 0.138
Inter-arrival[15] = 0.031
Inter-arrival[16] = 0.207
Inter-arrival[17] = 0.023
Inter-arrival[18] = 0.189
Inter-arrival[19] = 0.101
Inter-arrival[20] = 0.122
Inter-arrival[21] = 0.347
Inter-arrival[22] = 0.097
Inter-arrival[23] = 0.009
Inter-arrival[24] = 0.111
Inter-arrival[25] = 0.350
Inter-arrival[26] = 0.031
Inter-arrival[27] = 0.010
Inter-arrival[28] = 0.250
Inter-arrival[29] = 0.023
Inter-arrival[30] = 0.045
Inter-arrival[31] = 0.013
Inter-arrival[32] = 0.025
Inter-arrival[33] = 0.043
Inter-arrival[34] = 0.519
Inter-arrival[35] = 0.136
Inter-arrival[36] = 0.011
Inter-arrival[37] = 0.069
Inter-arrival[38] = 0.055
Inter-arrival[39] = 0.069
Inter-arrival[40] = 0.015
Inter-arrival[41] = 0.015
Inter-arrival[42] = 0.188
Inter-arrival[43] = 0.035
Inter-arrival[44] = 0.009
Inter-arrival[45] = 0.006
Inter-arrival[46] = 0.003
Inter-arrival[47] = 0.012
Inter-arrival[48] = 0.017
Inter-arrival[49] = 0.193
Inter-arrival[50] = 0.033
Inter-arrival[51] = 0.021
Inter-arrival[52] = 0.398
```

```
Inter-arrival[53] = 0.141
Inter-arrival[54] = 0.237
Inter-arrival[55] = 0.191
Inter-arrival[56] = 0.015
Inter-arrival[57] = 0.216
Inter-arrival[58] = 0.068
Inter-arrival[59] = 0.227
Inter-arrival[60] = 0.036
Inter-arrival[61] = 0.082
Inter-arrival[62] = 0.054
Inter-arrival[63] = 0.075
Inter-arrival[64] = 0.013
Inter-arrival[65] = 0.186
Inter-arrival[66] = 0.043
Inter-arrival[67] = 0.274
Inter-arrival[68] = 0.000
Inter-arrival[69] = 0.021
Inter-arrival[70] = 0.156
Inter-arrival[71] = 0.050
Inter-arrival[72] = 0.013
Inter-arrival[73] = 0.367
Inter-arrival[74] = 0.030
Inter-arrival[75] = 0.015
Inter-arrival[76] = 0.179
Inter-arrival[77] = 0.041
Inter-arrival[78] = 0.145
Inter-arrival[79] = 0.081
Inter-arrival[80] = 0.229
Inter-arrival[81] = 0.079
Inter-arrival[82] = 0.039
Inter-arrival[83] = 0.116
Inter-arrival[84] = 0.006
Inter-arrival[85] = 0.132
Inter-arrival[86] = 0.001
Inter-arrival[87] = 0.012
Inter-arrival[88] = 0.029
Inter-arrival[89] = 0.104
Inter-arrival[90] = 0.070
Inter-arrival[91] = 0.205
Inter-arrival[92] = 0.011
Inter-arrival[93] = 0.023
Inter-arrival[94] = 0.062
Inter-arrival[95] = 0.243
95
The expected value of interaarival time is 0.10
Inter-arrival times written to inter_arrival.txt
```

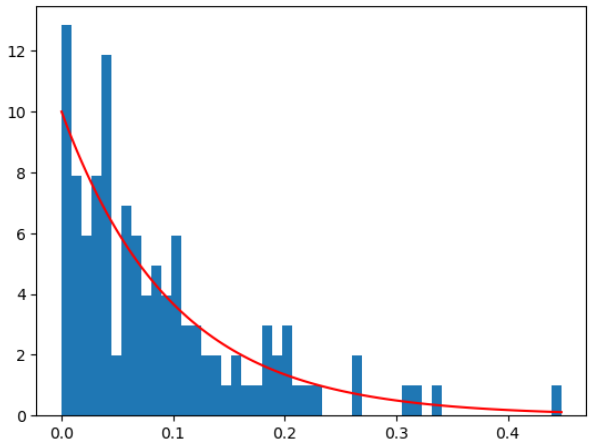


Figure 3: Inter arrival Time Modelling

DISCUSSION:
The experiment helped in demonstrating that a Poisson arrival process produces exponentially distributed inter-arrival times. The simulated mean inter-arrival time closely matched the calculated theoretical value of $1/\lambda$, therefore validating the model. Due to randomness, some deviations were observed. The histogram of inter-arrival times showed the expected exponential distribution which is right skewed.

ACKNOWLEDGEMENT:
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References

[1] E. Modiano, *6.263/16.37: Lectures 5 & 6 – Introduction to Queueing Theory*, Lecture slides, Cambridge, MA, USA.

[2] S. Chan, *ECE 302: Lecture 4.6 – Exponential Random Variable*, Lecture slides, West Lafayette, IN, USA, 2022.

APPENDIX (CODE):

Djikstra's Algorithm in C

```
#include <stdio.h>
#include <stdlib.h>
#include <math.h>
#include <time.h>

#define MAX_EVENTS 1000
float inter_arrival[MAX_EVENTS];
int ia_idx = 0;
// generating exponential RV
```

```
float exponential(float rate) {
    float u = (float)rand() / RAND_MAX;
    return -log(1.0 - u) / rate;
}

float expectation(float arr[], int n){

    float sum_ia = 0.0;
    for(int i = 0; i < n; i++){
        sum_ia += arr[i];
    }
    return sum_ia / n;

}
int main() {

    srand(time(NULL));

    // parameters
    float lambda = 10.0;     // arrival rate
    float mu = 13.0;         // service rate
    int K = 5;               // finite buffer size
    float SIM_TIME = 10.0;  // total simulation time

    // variables
    int buffer = 0;
    int arrivals = 0;
    int dropped = 0;
    int serviced = 0;

    // event times
    float t = 0.0;
    float next_arrival = exponential(lambda);
    float next_service = INFINITY;
    float arrival_time[MAX_EVENTS];
    //float inter_arrival[1000];
    int arr_idx = 0;
    //int ia_idx = 0;

    // loop being less than the total simulation timew
    while (t < SIM_TIME) {

        if (next_arrival < next_service) {
            // arrival process
            t = next_arrival;
            arrival_time[arr_idx++] = t;
            arrivals++;

            if (buffer < K) {
                buffer++;

                //If server is idle, schedule service
                if (buffer == 1) {
                    next_service = t + exponential(mu);
                }
            } else {
                dropped++;
            }
```

```
68
69              next_arrival = t + exponential(lambda);
70
71          } else {
72              // service
73              t = next_service;
74
75              if (buffer > 0) {
76                  buffer--;
77                  serviced++;
78              }
79
80              if (buffer > 0) {
81                  next_service = t + exponential(mu);
82              } else {
83                  next_service = INFINITY;
84              }
85          }
86          //printf("Event Time next arrival : %0.2f\n", t);
87          //printf("----------------------------------\n");
88      }
89
90
91
92      // result
93      printf("Simulation Time           : %.2f\n", SIM_TIME);
94      printf("Arrival rate (lambda)     : %.2f\n", lambda);
95      printf("Service rate (mu)         : %.2f\n", mu);
96      printf("Buffer size (K)           : %d\n", K);
97      printf("----------------------------------\n");
98      printf("Total Arrivals            : %d\n", arrivals);
99      printf("Total Serviced            : %d\n", serviced);
100     printf("Total Dropped             : %d\n", dropped);
101     printf("Packet Drop Probability   : %f\n", (float)dropped / arrivals);
102     printf("----------------------------------\n");
103
104
105     printf("Storing Interaarival Times\n");
106     for (int i = 1; i < arr_idx; i++)
107     {
108         inter_arrival[ia_idx] = arrival_time[i] - arrival_time[i-1];
109         printf("Inter-arrival[%d] = %.3f\n", i, inter_arrival[ia_idx]);
110         ia_idx++;
111     }
112     printf("%d\n", ia_idx);
113     float expected_value = expectation(inter_arrival, ia_idx);
114     printf("The expected value of interaarival time is %0.2f\n",
        expected_value);
115
116     FILE *fp = fopen("inter_arrival.txt", "w");
117
118     if (fp == NULL) {
119         printf("Error opening file!\n");
120         return 1;
121     }
122
123     for (int i = 0; i < ia_idx; i++) {
124         fprintf(fp, "%.6f\n", inter_arrival[i]);
```

```
    }

    fclose(fp);

    printf("Inter-arrival times written to inter_arrival.txt\n");

    return 0;
}
```

Listing 1: Python code for plotting the interarrival time

```
import numpy as np
import matplotlib.pyplot as plt

data = np.loadtxt("inter_arrival.txt")

plt.hist(data, bins=50, density=True)
t = np.linspace(0, max(data), 100)
plt.plot(t, 10*np.exp(-10*t), 'r')
plt.show()
```